

Measurement of the Rydberg Constant for Hydrogen

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1 Abstract

In this lab, the main goal was to get familiar with an optics device known as the Spencer Spectrometer. The spectrometer was used to measure diffraction angles from helium and hydrogen emission spectra. First, helium lines with known wavelengths were used to determine the grating spacing of the diffraction grating. This grating spacing was then used to calculate the wavelengths of the hydrogen Balmer lines. From these wavelengths, an experimental value of the Rydberg constant for hydrogen was found. The final measured value was

$$R = (1.0874 \pm 0.0073) \times 10^7 \text{ m}^{-1},$$

which had a percent error of about 0.91% compared to the accepted value.

2 Apparatus

- Spencer Spectrometer
- Diffraction Grating: 600 lines/mm
- Gas Discharge Tubes
- Power Supply
- Auxiliary white light source
- Aluminum Mirror

3 Procedure

3.1 Adjusting the Spectrometer

This lab consisted of two parts over the course of two weeks. For the first week, since the spectrometer was an unfamiliar device, we carefully got comfortable

with it and learned how to adjust it. To prepare the spectrometer for proper use, we had to learn how to do the following: focus the telescope for parallel light rays, focus the collimator for parallel rays, adjust the collimator and telescope axes to be perpendicular to the bearing axis, and then adjust the dispersive element.

Parallel rays are needed for the telescope because we need the image that is being produced in the focal plane to be clear. The collimator needs to be adjusted so that the light entering the dispersive element is correctly oriented, resulting in wavelengths being dispersed at well-defined angles. Once the spectrometer was properly adjusted, it was ready to make measurements of the helium and hydrogen spectra. Another potential issue for the spectrometer that needed to be resolved was parallax, and the autocollimation method was used to fix this and to get a better resolution of the image.

3.2 Making Measurements – Part 1

In Part 1, the helium spectrum was measured first. Since the wavelengths of strong helium lines are known, these lines were used as standards to experimentally determine the grating spacing d . The diffraction grating equation used was

$$d \sin \theta = m\lambda, \quad (1)$$

where d is the grating spacing, θ is the diffraction angle measured from the central maximum, m is the diffraction order, and λ is the wavelength of the spectral line.

To estimate d , this equation was rearranged into the linear form

$$m\lambda = d \sin \theta. \quad (2)$$

This has the form

$$y = dx,$$

where

$$y = m\lambda$$

and

$$x = \sin \theta.$$

A least-squares linear fit was then used, where the slope of the graph represents the grating spacing d .

Angles were measured using the vernier scale on the spectrometer. The main scale was measured in degrees and the vernier scale was measured in arcminutes. Since the vernier scale could be read to about $1'$, the uncertainty in each angular reading was taken to be about

$$\pm 0.5' = \pm 0.0083^\circ.$$

This uncertainty estimate also connects to the lab prompt about estimating uncertainty when a spectral line has a complex structure or is only partly resolved. In those cases, the uncertainty is not only from the vernier scale. It also depends on how clearly the center of the spectral line can be identified. If a line was broad, blurry, or partially resolved, then the uncertainty in centering the crosshairs on the line would be larger than the basic $\pm 0.5'$ reading uncertainty.

3.3 Making Measurements – Part 2

In Part 2, the hydrogen spectrum was measured. The grating spacing found from the helium spectrum was used to calculate the wavelengths of the hydrogen spectral lines. The same diffraction grating equation was used:

$$\lambda = \frac{d \sin \theta}{m}. \quad (3)$$

These calculated wavelengths were then compared to the standard values for the Balmer series of hydrogen. The hydrogen Balmer lines correspond to transitions where the electron falls down to the $n_f = 2$ energy level. The lines used in this experiment were

$$H_\alpha : n_i = 3,$$

$$H_\beta : n_i = 4,$$

and

$$H_\gamma : n_i = 5.$$

The Rydberg constant was calculated using

$$\frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{n_i^2} \right). \quad (4)$$

Solving for R ,

$$R = \frac{\frac{1}{\lambda}}{\frac{1}{4} - \frac{1}{n_i^2}}. \quad (5)$$

Here, n_i represents the initial energy level of the electron, while the final energy level is $n_f = 2$. Since the Rydberg formula predicts wavelengths in vacuum, the measured air wavelengths were converted to vacuum wavelengths using

$$\lambda_{\text{vacuum}} = n_{\text{air}} \lambda_{\text{air}}, \quad (6)$$

where

$$n_{\text{air}} = 1.00029.$$

This addresses the note at the end of Part 2 about air wavelengths versus vacuum wavelengths. The measured wavelengths and the visible NIST wavelengths are air wavelengths, while the Rydberg formula predicts vacuum wavelengths. For this reason, the wavelengths were converted before calculating R .

4 Analysis

4.1 Determination of the Grating Spacing

The first part of the analysis was to determine the grating spacing d using the helium data. The known helium wavelengths were used as standards, and the measured diffraction angles were used to make a least-squares fit of $m\lambda$ versus $\sin \theta$.

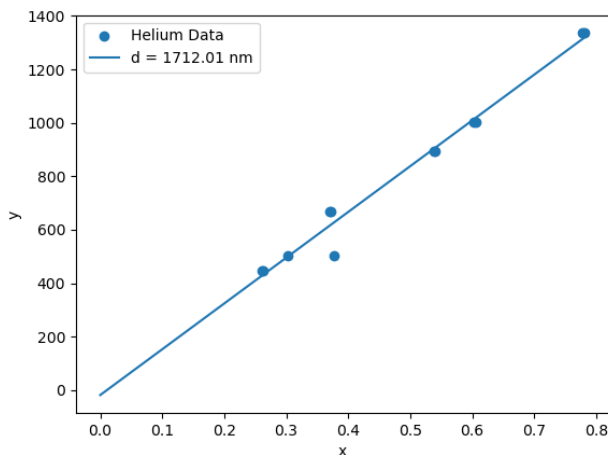


Figure 1: Least-squares fit used to determine the grating spacing d from the helium spectrum.

From the slope of the linear fit, the grating spacing was found to be

$$d = 1712.01 \text{ nm.}$$

This corresponds to a grating line density of

$$\frac{10^6}{1712.01} = 584 \text{ lines/mm.}$$

The diffraction grating was labeled as 600 lines/mm. The percent error in the measured line density was therefore

$$\% \text{ error} = \frac{|584 - 600|}{600} \times 100\% = 2.72\%.$$

This is reasonably close, though not exact. Some error likely came from uncertainty in centering the crosshairs on the spectral lines, the finite width of the spectral lines, and possible small errors in the alignment of the diffraction grating.

When different diffraction orders overlapped, the order of each observed line was determined by checking whether the measured angle was consistent with the grating equation. This addresses the lab prompt asking how the order of each observed line was deduced when spectra from different orders overlapped. Since larger orders appear at larger diffraction angles for the same wavelength, the measured angle could be compared with the expected value from

$$m\lambda = d \sin \theta.$$

This made it possible to identify whether a line was first order, second order, or another visible order. If an order assignment gave a value of d close to the fitted value of 1712.01 nm, then that order assignment was considered consistent. The spectrum did not appear perfectly symmetric in brightness on both sides of the zeroth order. This addresses the lab question asking whether one side of the spectrum seemed brighter than the other. The right side appeared somewhat brighter, which could happen because the diffraction grating may not have been perfectly aligned, and the incident light may not have hit the grating at exactly normal incidence. Small imperfections in the grating or uneven illumination of the slit could also make one side appear brighter.

4.2 Measurement of Hydrogen Wavelengths

Using the measured grating spacing from Part 1, the wavelengths of the hydrogen spectral lines were calculated. This table addresses the lab prompt asking to calculate the hydrogen wavelengths using the grating spacing from Part 1 and compare them with standard values. The measured wavelengths were compared with the standard air wavelengths for the Balmer series.

Spectral Line	Measured Wavelength (nm)	Standard Wavelength (nm)	Percent Error
H_γ	436.23	434.046	0.5%
H_β	491.92	486.135	1.2%
H_α	663.90	656.285	1.2%

Table 1: Measured hydrogen wavelengths compared with standard Balmer wavelengths.

The measured wavelengths were all larger than the standard values. This likely came from the measured grating spacing being larger than the nominal spacing. Since the wavelength was calculated using

$$\lambda = \frac{d \sin \theta}{m},$$

a value of d that is too large would also make the calculated wavelengths too large.

4.3 Calculation of the Rydberg Constant

After finding the hydrogen wavelengths, the Rydberg constant was calculated for each spectral line using

$$R = \frac{\frac{1}{\bar{\lambda}}}{\frac{1}{4} - \frac{1}{n_i^2}}.$$

Before calculating R , the wavelengths were converted from air wavelengths to vacuum wavelengths using

$$\lambda_{\text{vacuum}} = 1.00029 \lambda_{\text{air}}.$$

The average experimental value found for the Rydberg constant was

$$R = 1.0874307 \times 10^7 \text{ m}^{-1}.$$

The uncertainty was estimated using the standard error of the mean of the individual Rydberg constant values:

$$\sigma_{\bar{R}} = \frac{\sigma_R}{\sqrt{N}}.$$

This gave

$$\sigma_{\bar{R}} = 7.2812 \times 10^4 \text{ m}^{-1}.$$

Therefore, the final measured value was

$$R = (1.0874 \pm 0.0073) \times 10^7 \text{ m}^{-1}.$$

The accepted value of the Rydberg constant is approximately

$$R = 1.097 \times 10^7 \text{ m}^{-1}.$$

The percent error was

$$\% \text{ error} = \frac{|1.0874307 \times 10^7 - 1.097 \times 10^7|}{1.097 \times 10^7} \times 100\% \approx 0.91\%.$$

This is a fairly reasonable result for this experiment. The largest sources of error likely came from uncertainty in reading the angular positions, imperfect alignment of the spectrometer, and the fact that some spectral lines were difficult to center precisely with the crosshairs.

5 Conclusion

In this experiment, the Spencer Spectrometer was used to measure the emission spectra of helium and hydrogen. The helium spectrum was first used to determine the grating spacing of the diffraction grating. A least-squares fit gave

$$d = 1712.01 \text{ nm},$$

which corresponds to about

$$584 \text{ lines/mm}.$$

This had a percent error of 2.72% compared to the nominal value of 600 lines/mm.

The grating spacing was then used to calculate the wavelengths of the hydrogen Balmer lines. The measured values were close to the standard wavelengths, with percent errors of about 0.5% to 1.2%. Finally, these wavelengths were used to calculate the Rydberg constant. The experimental value was

$$R = (1.0874 \pm 0.0073) \times 10^7 \text{ m}^{-1},$$

which had a percent error of about 0.91%. Overall, the result was reasonably close to the accepted value, showing that the spectrometer and diffraction grating can be used to measure atomic emission wavelengths and estimate the Rydberg constant.